

About This Course

Several teachers from the THz Sensing Group:

A. Neto, D. Cavallo, N. LLombart

Preparation for Master Thesis topics

All about understanding and analyzing high frequency antennas

Lots of Theory and Programming:

Structure

Per week: 2h theory / 2h Matlab instruction

Every week there are Matlab assignments (focus on programming)

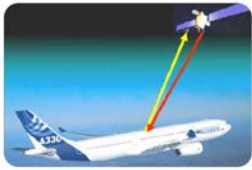
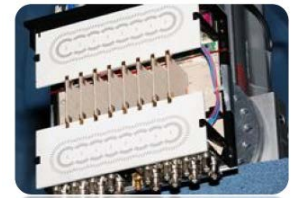
Final Mark: Assignments + Final individual presentation (date to be decided)

Examples of Individual Projects

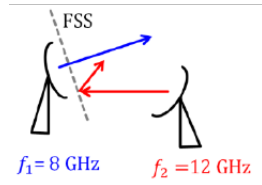
Design a dipole antenna integrated on a chip



Design a linear array of dipoles for a FMCW radar front-end



Design an antenna array to be integrated onto an airplane for aircraft-satellite communication

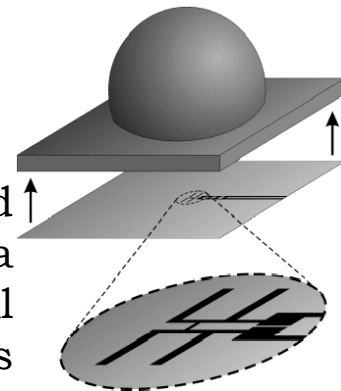


Design an FSS radome for a military ship



Design a reflector antenna for a satellite telecom application at 30GHz feed.

Design an integrated quasi-optical antenna for astronomical detectors



Schedule

			Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Tuesday	15:45 - 17:45	EWI-Lecture hall F						EUCAP	
			Spectral GF	Reflectors	Lenses	FO/FPA	Phased Array		Phased Array
			A. Neto	A. Neto	N. Llombart	N. Llombart	D. Cavallo		D. Cavallo
Thursday	13:45 - 15:45	EWI-Lecture hall G							
			Matlab	Matlab	Matlab	Matlab	Matlab		Matlab

ET4580 Quasi Optical Systems

Topic 1

Free Space Spectral Green's Function

Learning Objectives

Free Space Green's function

Spatial Fourier Transforms

Asymptotic evaluation of integrals via saddle point

Green's functions

$\bar{\bar{G}}(\vec{r} - \vec{r}')$
 A Green's function is
 The Field, $\bar{\bar{G}}$, in $\vec{r}$
 radiated by an elementary $\delta(\vec{r} - \vec{r}')\hat{p}$
 equivalent current source,
 in $\vec{r}'$

In the case that one investigates
electric field radiated by **electric currents**

$$\vec{e}(\vec{r}) = \iiint_V \bar{\bar{g}}^{ej}(\vec{r} - \vec{r}') * j(\vec{r}') d\vec{r}'$$

Convolution integral between
 Green's Functions and
 Equivalent currents describes
 the Fields

These representation is virtually the only one
 that allows you to solve a non trivial problem

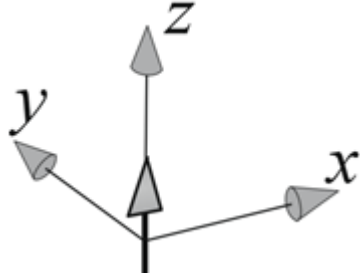
Green's function of electric current in Free Space

From the magnetic potential A

Potential are a tool invented to rendered easier the derivations of Green's functions

$$\vec{e}(\vec{r}) = -j\omega\vec{A} - \frac{j}{\omega\epsilon\mu} \nabla[\nabla \cdot \vec{A}]$$

$$\vec{h}(\vec{r}) = \frac{1}{\mu} \nabla \times \vec{A}$$

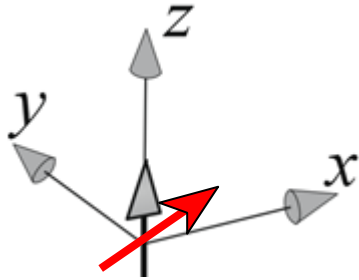
$$\vec{j}(\vec{r}) = \delta(x)\delta(y)\delta(z)\hat{z}$$




$$\left\{ \begin{array}{l} \lim_{r \rightarrow \infty} \vec{h}(\vec{r}) = jk \frac{\sin\theta e^{-jkr}}{4\pi r} \hat{\phi} \\ \lim_{r \rightarrow \infty} \vec{e}(\vec{r}) = jk\zeta \frac{\sin\theta e^{-jkr}}{4\pi r} \hat{\theta} \end{array} \right.$$

$$\vec{A}(\vec{r}, \vec{j}) = \frac{\mu e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} \hat{z}$$

Free Space Dyadic Green's function of electric current in Free Space



For dipoles not along z

$$\vec{j}(\vec{r}) = \delta(x)\delta(y)\delta(z)\hat{p}$$

$$\vec{A}(\vec{r}, \vec{j}) = \frac{\mu e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} \hat{p}$$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = -j \frac{\zeta}{k} \begin{vmatrix} k^2 + \frac{\partial^2}{\partial x^2} & \frac{\partial^2}{\partial x \partial y} & \frac{\partial^2}{\partial x \partial z} \\ \frac{\partial^2}{\partial y \partial x} & k^2 + \frac{\partial^2}{\partial y^2} & \frac{\partial^2}{\partial y \partial z} \\ \frac{\partial^2}{\partial z \partial x} & \frac{\partial^2}{\partial z \partial y} & k^2 + \frac{\partial^2}{\partial z^2} \end{vmatrix} \frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|}$$

The Green's function is not analytical because the derivatives, which arise from the potentials, still need to be performed.

Spectral Representations of Green's Function

- An alternative expression for the electric field Green's function. We will present the GF in its **spectral form**.

$$G(k_x) = \int_{-\infty}^{\infty} g(x) e^{jk_x x} dx \qquad g(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(k_x) e^{-jk_x x} dk_x$$

This is the one dimensional FT couple we will use

- Thus not a completely analytical form, we will still have to perform an **inverse Fourier transform** to obtain a space domain expression.
- It might look like a step back, but in practice this expression can be extremely useful to treat complex configurations, closer to real applications, in which closed form space domain expressions do not exist.

Representations of Dirac's Delta

The Fourier Transform of the Delta $FT\{\delta(x - x')\}$

$$FT\{\delta(x - x')\} = \int_{-\infty}^{\infty} \delta(x - x') e^{jk_x x} dx = e^{jk_x x'}$$

Taking the inverse FT

$$FT^{-1}\{e^{jk_x x'}\} = \frac{1}{(2\pi)} \int_{-\infty}^{\infty} e^{jk_x x'} e^{-jk_x x} dk_x$$
$$= \frac{1}{(2\pi)} \int_{-\infty}^{\infty} e^{-jk_x (x - x')} dk_x = \delta(x - x')$$


Spectral representation
of Dirac's Delta function

Representations of vectorial Dirac's Delta

$$\delta(\vec{r} - \vec{r}') = \delta(x)\delta(y)\delta(z)$$

$$\begin{aligned} \delta(\vec{r} - \vec{r}') \\ = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z \end{aligned}$$

Spectral representation of 3D Dirac's Delta function
in **cartesian coordinates**

Spectral Representation of the Scalar GF

$$\nabla^2 \psi + k^2 \psi = -\delta(\vec{r} - \vec{r}')$$

Helmoltz equation for potential

$$\psi = \frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r} - \vec{r}'|}$$

is the solution, after imposing the radiation condition

We can use this information/definition to obtain a spectral representation of ψ

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r} - \vec{r}'|} = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Psi(k_x, k_y, k_z) e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z$$

This means deriving the expression of $\Psi(k_x, k_y, k_z, \vec{r}')$

The starting point is the representation of the 3D delta function in cartesian coordinates

$$\delta(\vec{r} - \vec{r}') = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z$$

Spectral Representation of the Scalar GF (2)

$$\nabla^2 \psi + k^2 \psi = -\delta(\vec{r} - \vec{r}')$$

If we then use the spectral representation of both sides of the Helmholtz equation

$$\begin{aligned} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [\nabla^2 + k^2] \Psi(k_x, k_y, k_z, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \\ = -\frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z \end{aligned}$$

Performing the algebraic derivatives on the LHS

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial^2 x} + \frac{\partial^2}{\partial^2 y} + \frac{\partial^2}{\partial^2 z}$$



In spectrum is $-k_x^2 - k_y^2 - k_z^2$

$$\begin{aligned} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [k^2 - k_x^2 - k_y^2 - k_z^2] \Psi(k_x, k_y, k_z, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \\ = -\frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{jk_x x'} e^{jk_y y'} e^{jk_z z'} e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \end{aligned}$$

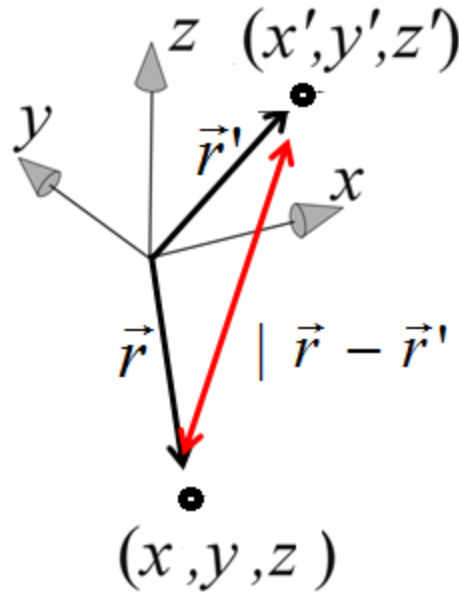
Spectral Representation of the Scalar GF (3)

$$\begin{aligned} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [k^2 - k_x^2 - k_y^2 - k_z^2] \Psi(k_x, k_y, k_z, \vec{r}') e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \\ = -\frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{jk_x x'} e^{jk_y y'} e^{jk_z z'} e^{-jk_x x} e^{-jk_y y} e^{-jk_z z} dk_x dk_y dk_z \end{aligned}$$

Equating the spectra since the equation is valid for every (x,y,z)

$$\Psi(k_x, k_y, k_z, \vec{r}') = -\frac{e^{jk_x x'} e^{jk_y y'} e^{jk_z z'}}{k^2 - k_x^2 - k_y^2 - k_z^2}$$

Spectral Representation of the Scalar GF (4)



$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} -\frac{e^{-jk_x(x-x')}e^{-jk_y(y-y')}e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

Spectral Identity and Derivatives

The spectral representation of the scalar potential leads to easy derivatives for the GF

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} -\frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

$$\frac{\partial^2}{\partial x^2} \rightarrow -k_x^2; \quad \frac{\partial^2}{\partial x \partial y} = -k_x k_y; \quad \frac{\partial^2}{\partial x \partial z} = -k_x k_z$$

$$\frac{\partial^2}{\partial y \partial x} \rightarrow -k_x k_y; \quad \frac{\partial^2}{\partial y^2} = -k_y^2; \quad \frac{\partial^2}{\partial y \partial z} = -k_y k_z$$

$$\frac{\partial^2}{\partial z \partial x} \rightarrow -k_z k_x; \quad \frac{\partial^2}{\partial z \partial y} = -k_z k_y; \quad \frac{\partial^2}{\partial z^2} = -k_z^2$$

Spectral Representation of the GF

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{vmatrix} k^2 - k_x^2 & -k_x k_y & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{vmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

- The derivatives are analytical but the integrals have to be performed

2D integral representation

For any $(z - z') \neq 0$

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = -j \frac{1}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2-k_x^2-k_y^2}|z-z'|}}{\sqrt{k^2-k_x^2-k_y^2}} dk_x dk_y$$

Demonstration

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = -\frac{1}{8\pi^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-jk_x(x-x')} e^{-jk_y(y-y')} \int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{k_p^2 - k_z^2} dk_x dk_y$$

where

$$k^2 - k_x^2 - k_y^2 = k_p^2$$

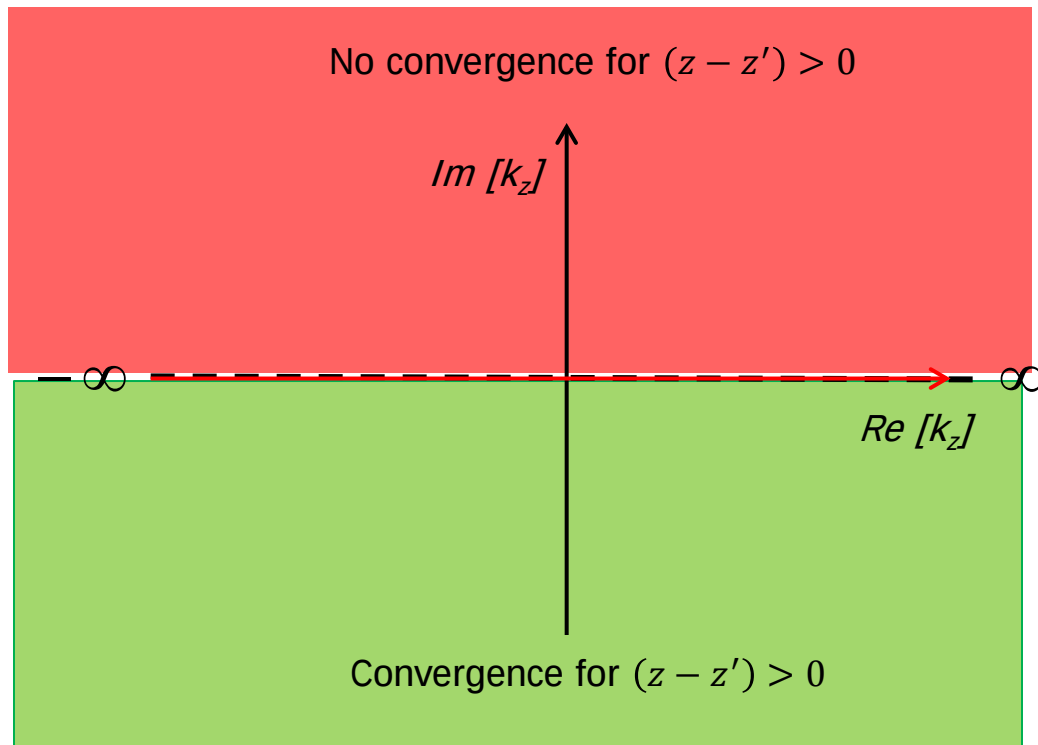
And the using

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z = \pi j \frac{e^{-jk_p|z-z'|}}{k_p}$$

Substituting in the previous the demonstration is finished

Complex Plane Convergence

$$(z - z') > 0$$



$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p^2 - k_z^2)} dk_z$$

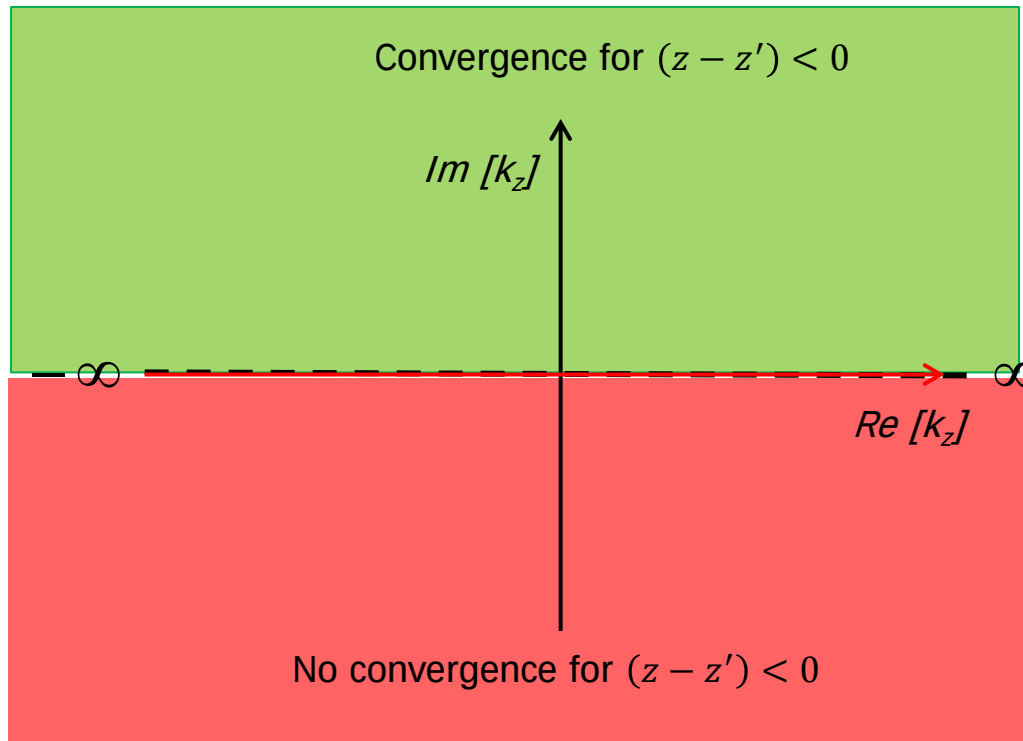
for $|k_z| \rightarrow \infty \left| \frac{1}{(k_p^2 - k_z^2)} \right| \rightarrow 0$

On real axis $e^{-jk_z(z-z')}$ remains bounded

$$\lim_{Im(k_z) \rightarrow \infty} |e^{-jk_z|z-z'|}| = |e^{-j(j|k_z|)|z-z'|}| = |e^{|k_z||z-z'|}| \rightarrow \infty$$

$$\lim_{Im(k_z) \rightarrow -\infty} |e^{-jk_z|z-z'|}| = |e^{-j(-j|k_z|)|z-z'|}| = |e^{-|k_z||z-z'|}| \rightarrow 0$$

$$(z - z') < 0$$



$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p^2 - k_z^2)} dk_z$$

for $|k_z| \rightarrow \infty \left| \frac{1}{(k_p^2 - k_z^2)} \right| \rightarrow 0$

On real axis $e^{-jk_z(z-z')}$ remains bounded

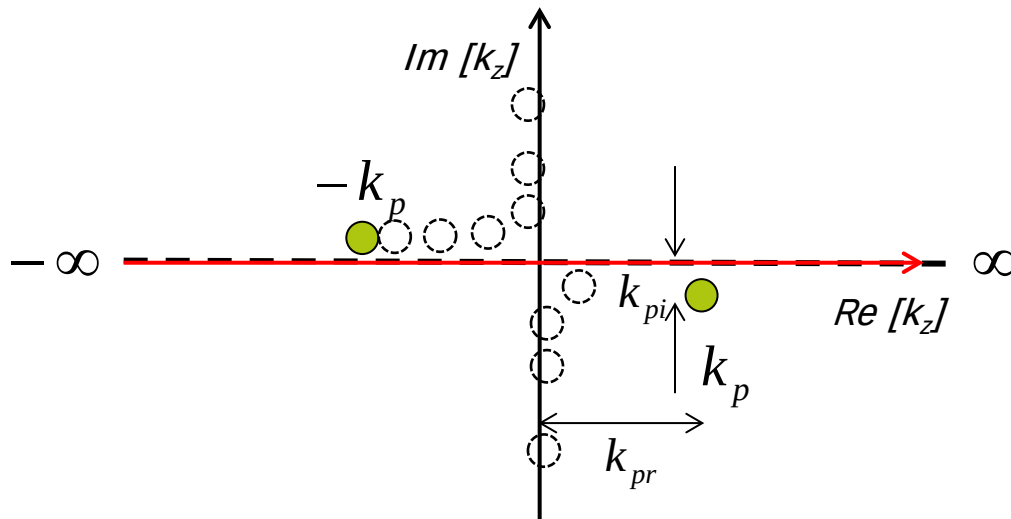
$$\lim_{Im(k_z) \rightarrow \infty} |e^{jk_z|z-z'|}| = |e^{j(j|k_z|)|z-z'|}| = |e^{-|k_z||z-z'|}| \rightarrow 0$$

$$\lim_{Im(k_z) \rightarrow -\infty} |e^{jk_z|z-z'|}| = |e^{j(-j|k_z|)|z-z'|}| = |e^{|k_z||z-z'|}| \rightarrow \infty$$

Complex Plane Topology

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z = \pi j \frac{e^{-jk_p|z-z'|}}{k_p}$$

Two poles



Why?

$$k_p = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{(k_r - jk_i)^2 - k_x^2 - k_y^2}$$

For

$$k_x = 0; \text{ and } k_y = 0$$

$$k_p = \sqrt{(k_r - jk_i)^2} = (k_r - jk_i)$$

$$k_p = k_{pr} - jk_{pi}$$

with

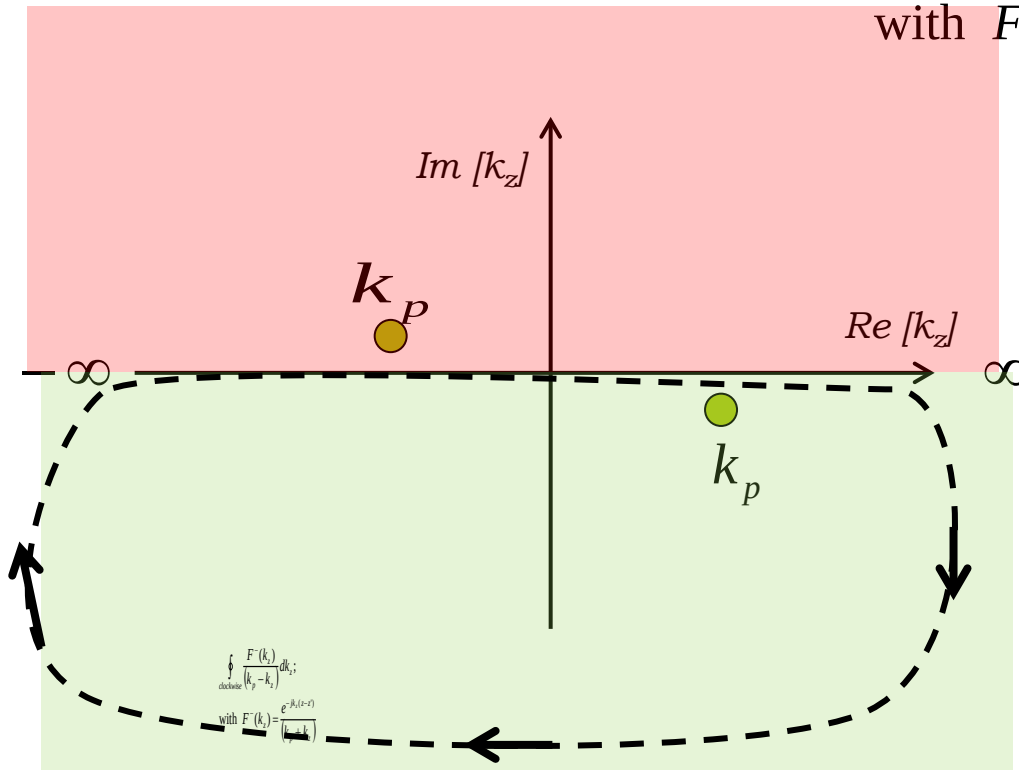
$$k_{pr} > 0 \text{ and } k_{pi} > 0$$

Spectral Identity

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z = \oint_{\text{clockwise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z;$$

Analytical function
in lower half space

with $F^-(k_z) = \frac{e^{-jk_z(z-z')}}{(k_p + k_z)}$



$$I_{\text{closed}} = I + I_{\text{lhs}}$$

$$I_{\text{lhs}} = 0$$

Because Integrand goes to zero at infinity

$$I_{\text{closed}} = I$$

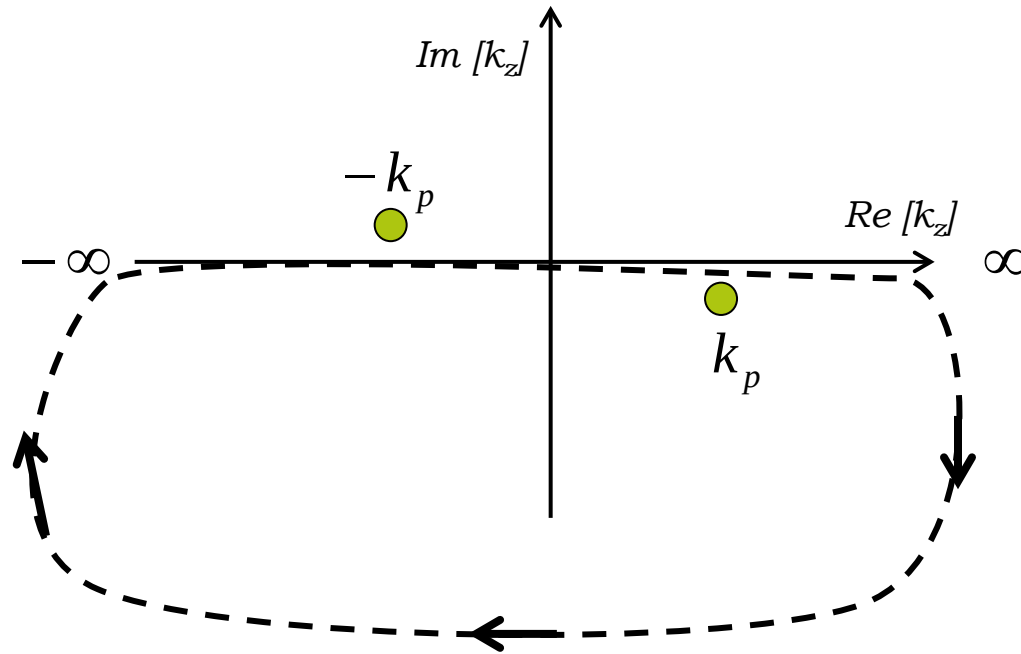
Circular Integral

$$I = \int_{\text{closed at } \infty}^{\text{clock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z$$

A closed integration
around a domain
with at least one polar singularity

Not simple as it is ...

However it can be expressed
in a simpler form

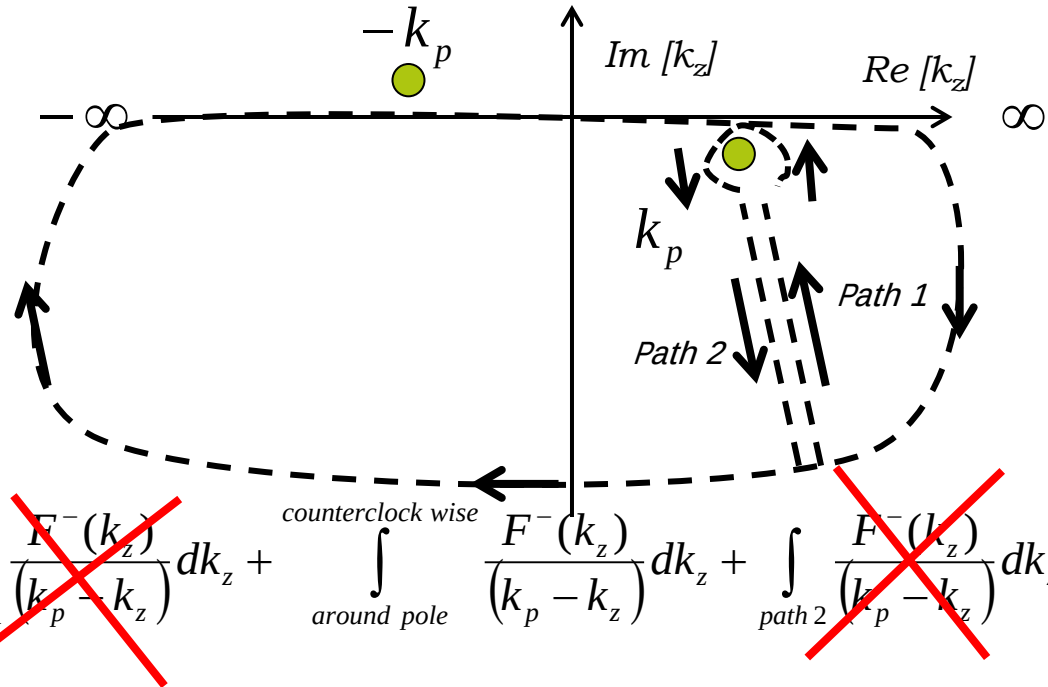


Cauchy Theorem

$$I = \int_{\text{closed at } \infty}^{\text{clock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z$$

$F^-(k_z)$ is analytical for $\text{Im}(k_z) < 0$

A closed integration around a domain with no singularities is zero



$$0 = \int_{\text{Closed at Infinity}}^{\text{clock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z + \cancel{\int_{\text{path 1}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z} + \int_{\text{around pole}}^{\text{counterclock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z + \cancel{\int_{\text{path 2}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z}$$

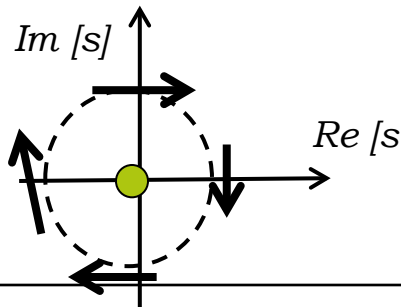
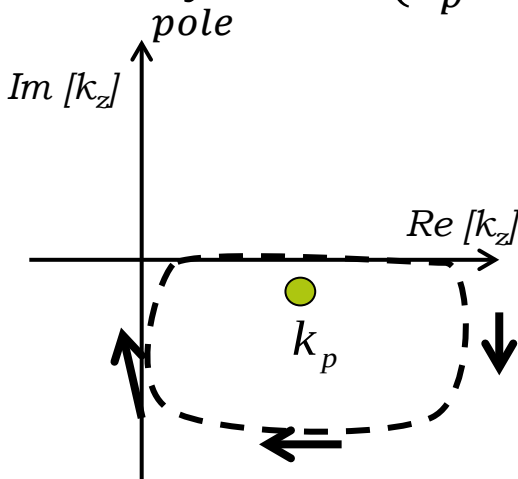
$$\int_{\text{Closed at Infinity}}^{\text{clock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z = - \int_{\text{around pole}}^{\text{counterclock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z$$

$$I = \int_{\text{around pole}}^{\text{clock wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z$$

Integration around a simple pole (clock wise)

If $F^-(k_z)$ is analytic function

$$I = \oint_{\text{clock-wise}} \frac{F^-(k_z)}{(k_p - k_z)} dk_z = 2\pi j F^-(k_p)$$



Step 1

$$(k_p - k_z) = s$$

$$I = -\oint \frac{F^-(k_p - s)}{s} ds$$

Step 2

$s = \rho e^{-j\phi}$ with ρ arbitrarily small

$$\begin{aligned} I &= -\lim_{\rho \rightarrow 0} \int_0^{2\pi} \frac{F^-(k_p - \rho e^{-j\phi})}{\rho e^{-j\phi}} \frac{ds}{d\phi} d\phi \\ &= -F^-(k_p) \int_0^{2\pi} \frac{1}{\rho e^{-j\phi}} (-j) \rho e^{-j\phi} d\phi \\ &= jF^-(k_p) \int_0^{2\pi} d\phi = 2\pi j F^-(k_p) \end{aligned}$$

Same steps for $z-z' < 0$

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z =$$

$$\oint_{\text{counterclockwise}} \frac{F^+(k_z)}{(k_p + k_z)} dk_z;$$

$$\text{with } F^+(k_z) = \frac{e^{-jk_z(z-z')}}{(k_p - k_z)}$$

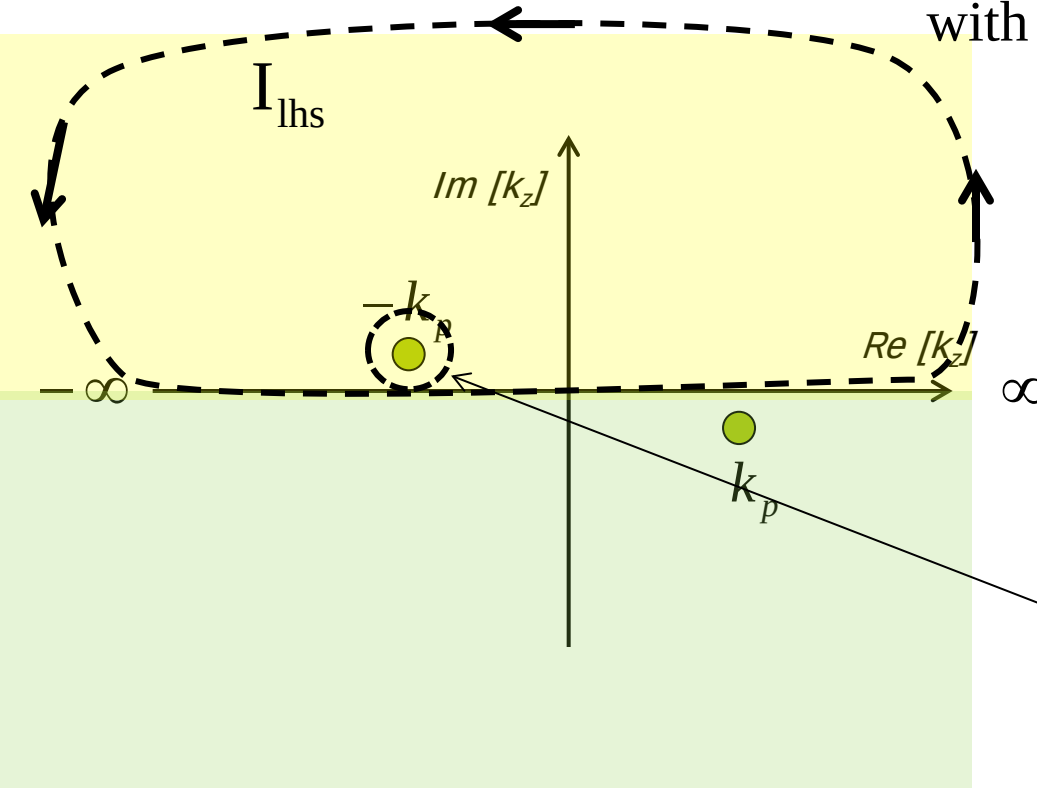
if $(z - z') < 0$

We can deform integral in upper half space to infinity

$$\begin{aligned} I_{\text{closed}} &= I + I_{\text{uhs}} \\ I_{\text{uhs}} &= 0 \end{aligned}$$

Because Integrand goes to zero at infinity

$$I_{\text{closed}} = \oint_{\text{counterclockwise}} \frac{F^+(k_z)}{(k_p + k_z)} dk_z$$



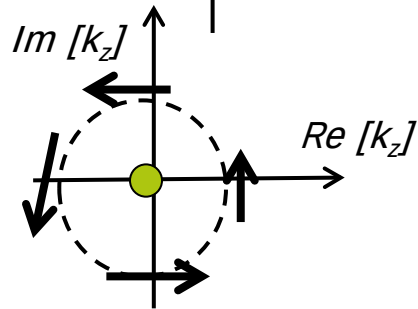
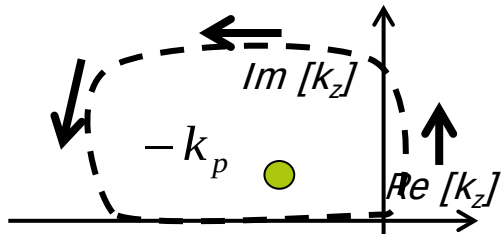
In complex domain you can deform integrals wherever there are no singularities

Integration around a simple pole

(counterclockwise)

If $F(k_z)$ is analytic function

$$I = \oint_{\text{counterclockwise}} \frac{F^+(k_z)}{(k_p + k_z)} dk_z = 2\pi j F^+(k_p)$$



Step 1
($k_p + k_z$) = s

$$I = \oint_{\text{counterclockwise}} \frac{F^+(s - k_p)}{s} ds$$

Step 2

$s = \rho e^{j\phi}$ with ρ arbitrarily small

$$\begin{aligned} I &= \lim_{\rho \rightarrow 0} \int_0^{2\pi} \frac{F^+(\rho e^{j\phi} - k_p)}{\rho e^{j\phi}} \frac{ds}{d\phi} d\phi = F^+(-k_p) \int_0^{2\pi} \frac{1}{\rho e^{j\phi}} (j) \rho e^{j\phi} d\phi \\ &= jF^+(-k_p) \int_0^{2\pi} d\phi = 2\pi j F^+(-k_p) \end{aligned}$$

General Case

$$\text{if } (z - z') > 0 \quad I^- = 2\pi j F^-(k_p)$$

$$F^-(k_z) = \frac{e^{-jk_z(z-z')}}{(k_p + k_z)}$$

$$\text{if } (z - z') < 0 \quad I^+ = 2\pi j F^+(-k_p)$$

$$F^+(k_z) = \frac{e^{-jk_z(z-z')}}{(k_p - k_z)}$$

$$I^- = 2\pi j \left. \frac{e^{-jk_z(z-z')}}{(k_p + k_z)} \right|_{k_z=k_p} = \pi j \frac{e^{-jk_p(z-z')}}{k_p}$$

$$I^+ = 2\pi j \left. \frac{e^{-jk_z(z-z')}}{(k_p - k_z)} \right|_{k_z=-k_p} = \pi j \frac{e^{jk_p(z-z')}}{k_p}$$

$$\forall (z - z')$$

$$I^\pm = \pi j \frac{e^{-jk_p|z-z'|}}{k_p}$$

Spectral Identity

$$\int_{-\infty}^{\infty} \frac{e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z = j\pi \frac{e^{-jk_p|z-z'|}}{k_p}$$

For any $z-z' \neq 0$

Generalizing to analytical functions $C(k_z)$

where $\pm$ applies for $(z - z') \begin{matrix} > \\ < \end{matrix} 0$

$$\int_{-\infty}^{\infty} \frac{C(k_z)e^{-jk_z(z-z')}}{(k_p - k_z)(k_p + k_z)} dk_z = j\pi \frac{e^{-jk_p|z-z'|}}{k_p} C(\pm k_p)$$

Double Integral Representation of the GF

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{vmatrix} k^2 - k_x^2 & -k_x k_y & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{vmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

For any $(z - z') \neq 0$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = -\frac{\zeta}{k} \frac{1}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm$$

Spectral Representation of the GF when $\mathbf{z}=\mathbf{z}'$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r}-\vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{vmatrix} k^2 - k_x^2 & -k_x k_y & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{vmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r}-\vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2)}{k^2 - k_x^2 - k_y^2 - k_z^2} e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')} dk_x dk_y dk_z$$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r}-\vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left[\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2 + k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z \right] e^{-jk_x(x-x')} e^{-jk_y(y-y')} dk_y dk_x$$

Spectral Representation of the GF when $z=z'$

$$\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2 + k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z = \frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z + \frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{k_\rho^2}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z$$

$$\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2 + k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z = \frac{1}{(2\pi)} \int_{-\infty}^{\infty} e^{-jk_z(z-z')} dk_z + \frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{k_\rho^2}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z$$

$$\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2 + k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z = \delta(z - z') + \frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{k_\rho^2}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z$$

$$\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{k_\rho^2}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z = \frac{j}{2} \frac{k^2 - k_z^2}{\sqrt{k^2 - k_z^2}} e^{-j\sqrt{k^2 - k_z^2}|z-z'|}$$

$$\frac{1}{(2\pi)} \int_{-\infty}^{\infty} \frac{(k^2 - k_z^2 - k_\rho^2 + k_\rho^2)}{k^2 - k_\rho^2 - k_z^2} e^{-jk_z(z-z')} dk_z = \delta(z - z') + \frac{j}{2} \frac{k^2 - k_z^2}{\sqrt{k^2 - k_\rho^2}} e^{-j\sqrt{k^2 - k_\rho^2}|z-z'|}$$

Double Integral Representation of the GF

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{j\zeta}{k} \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{vmatrix} k^2 - k_x^2 & -k_x k_y & -k_x k_z \\ -k_y k_x & k^2 - k_y^2 & -k_y k_z \\ -k_z k_x & -k_z k_y & k^2 - k_z^2 \end{vmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

For any $(z - z') \neq 0$

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = -\frac{\zeta}{k} \frac{1}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & -2j\delta(z - z') + (k^2 - k_z^2) \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm$$

2D integrations

Typically we encounter always double rather than triple integral representations

They are mostly based on the mathematical identities like this

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = -j \frac{1}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-j\sqrt{k^2-k_x^2-k_y^2}|z-z'|}}{\sqrt{k^2-k_x^2-k_y^2}} dk_x dk_y$$

Nothing else than Fourier Transforms

Comments on Spectral GF

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{-\zeta}{k8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{k_z} dk_x dk_y$$

Key points:

- Observing these integral expression a few considerations can be made on the convergence.
- The diagonal terms of the dyad are composed of the sum of two terms.
 - The first term derives directly from the potential function.
 - While, second term and all the extra diagonal elements of the dyad are consequence of the derivations performed on the potential function.

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{-\zeta}{k8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{\bar{D}}(k_x, k_y) \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{k_z} dk_x dk_y$$

- These latter terms are characterized by a much slower convergence.
- In particular for $r = r'$ the convergence is not even achieved at all and the Greens function value tends to infinite.
- This should not surprise as the Greens function does not represent an observable quantity by itself but only when inserted in a convolution product.

Plane Wave representation of Fields

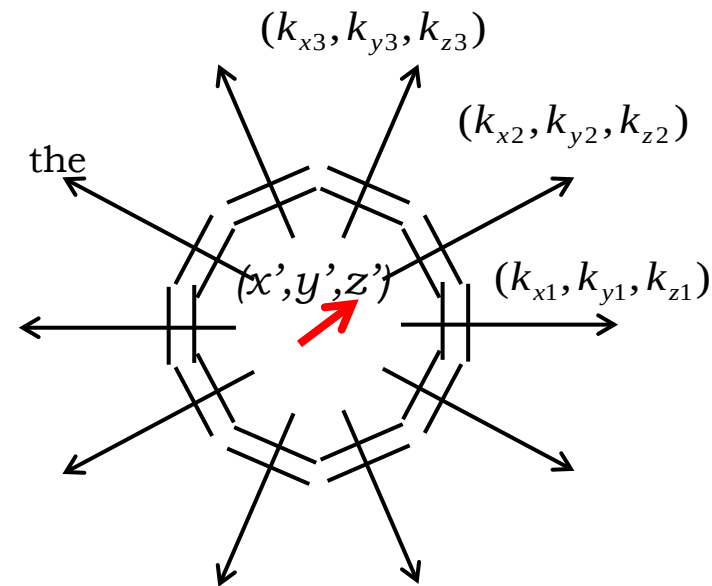
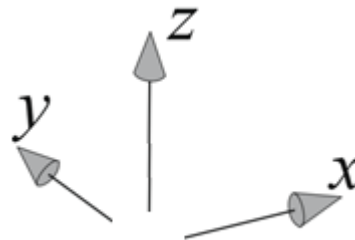
$$\bar{\bar{G}}_{fs}^{ej}(\vec{r} - \vec{r}') = \frac{-\zeta}{k8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm k_z = \sqrt{k^2 - k_x^2 - k_y^2}$$

This is the plane wave representation of E.M. Fields

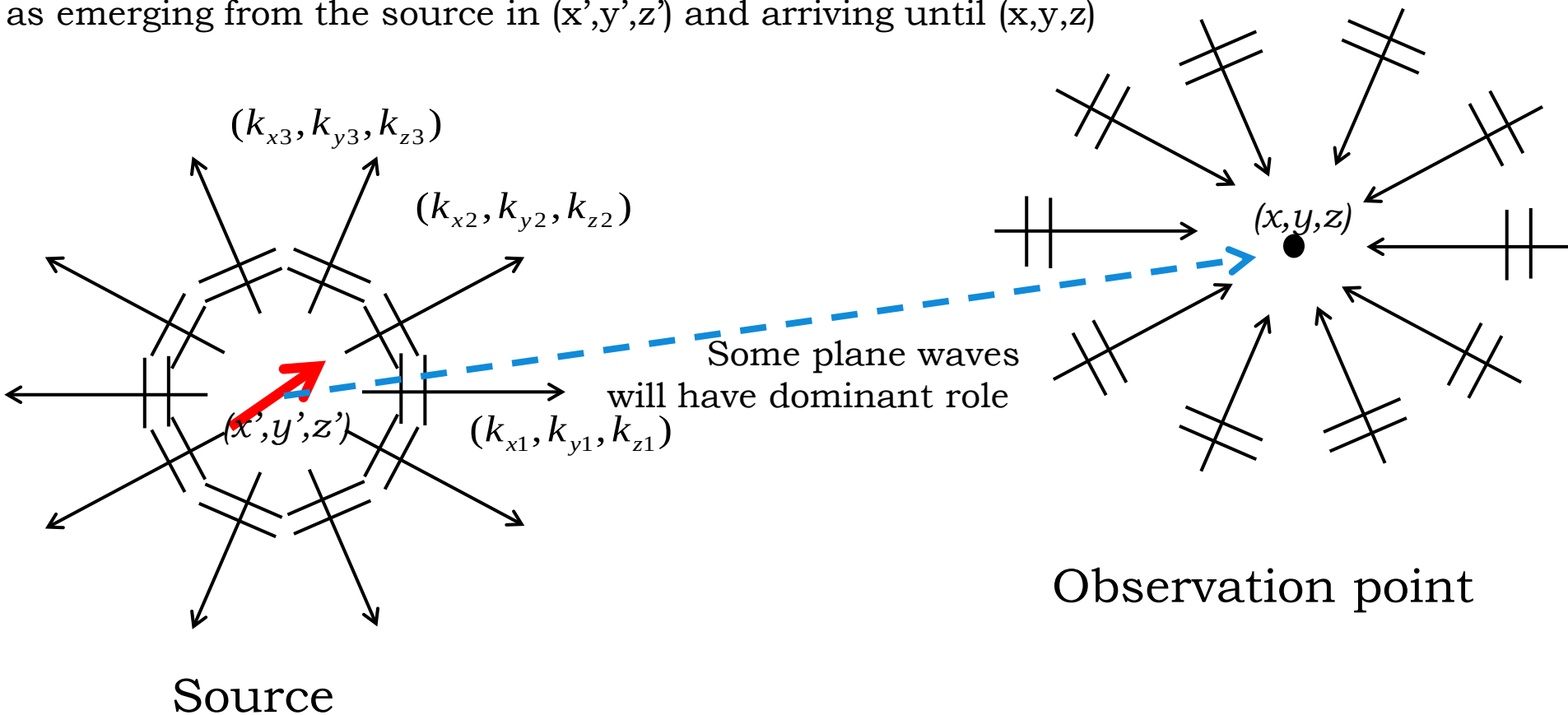
Because

- 1) the dependence from space variables is only at the exponent.
- 2) $k_x^2 + k_y^2 + k_z^2 = k^2$



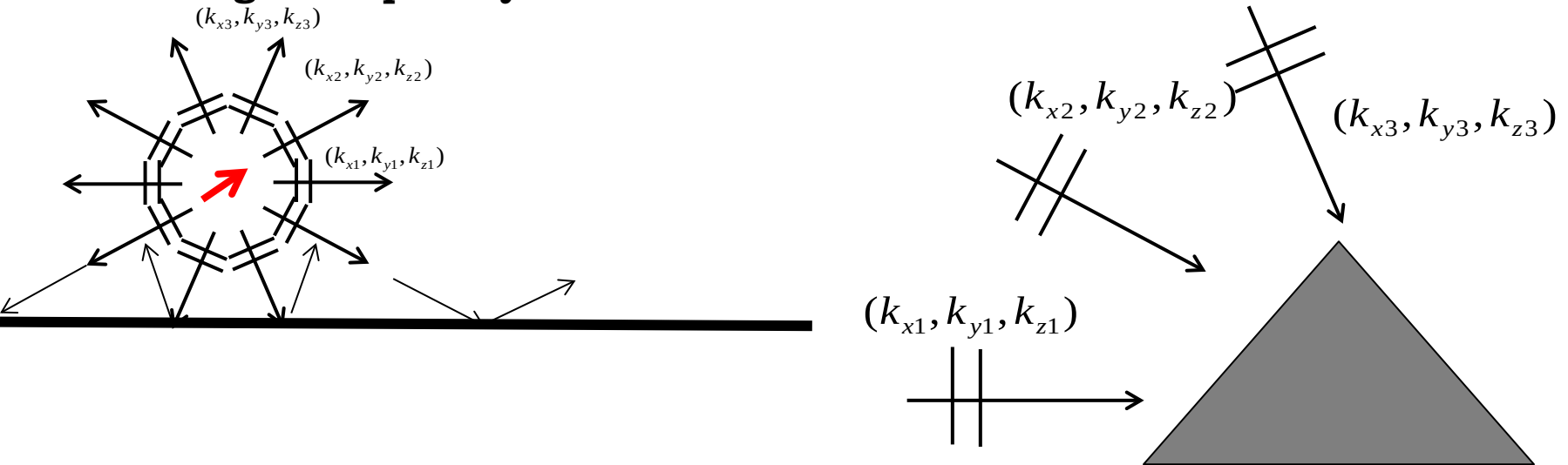
Plane Wave expansion

Plane waves are defined over the entire space.
Plane wave expansion represents the total radiated fields
as emerging from the source in (x',y',z') and arriving until (x,y,z)



Generic Obstacles

- Power of spectral GF comes from facility to build other Green's functions:
if you know response to a single plane waves,
than you can sum up all contributions to obtain response to elementary current
- **Infinite ground plane you will do as exercise**



Patch antennas are printed on dielectric slabs.

The GF in space domain is impossible but in spectral domain is much, much simpler

Magnetic Field due to Magnetic Currents

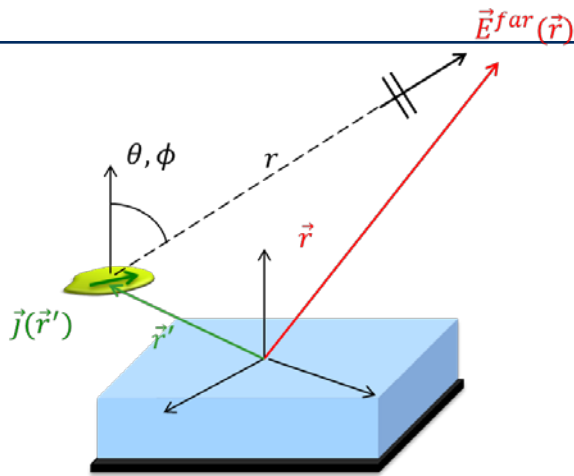
Proceeding in perfect analogy with the electric current case

$$\bar{\bar{G}}_{fs}^{hm}(\vec{r} - \vec{r}') = \frac{-1}{k\zeta 8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$(z - z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm \quad k_z = \sqrt{k^2 - k_x^2 - k_y^2}$$

Only difference is $\frac{1}{\zeta}$ instead of ζ

Radiation from current distributions



$$\vec{e}(x, y, z) = \int_{D_x} \int_{D_y} \bar{\bar{g}}^{ej}(x - x', y - y', z - z') \vec{j}(x', y') dx' dy'$$

$$\bar{\bar{g}}^{ej}(x - x', y - y', z - z') = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{\bar{G}}^{ej}(k_x, k_y) e^{-jk_x(x-x')} e^{-jk_y(y-y')} e^{-jk_z|z-z'|} dk_x dk_y$$

$$\vec{e}(x, y, z) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\bar{\bar{G}}^{ej}(k_x, k_y) k_z}{k_z} e^{-jk_x x} e^{-jk_y y} e^{-jk_z|z-z'|} \underbrace{\int_{D_x} \int_{D_y} e^{jk_x x'} e^{jk_y y'} \vec{j}(x', y') dx' dy'}_{\vec{J}(k_x, k_y)}$$

$$\bar{\bar{D}}^{ej}(k_x, k_y) \equiv \bar{\bar{G}}^{ej}(k_x, k_y) k_z$$

$$\vec{e}(x, y, z) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{\bar{D}}^{ej}(k_x, k_y) \vec{J}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z|z-z'|}}{k_z} dk_x dk_y$$

Far Field Radiation

$$\vec{f}(x, y, z) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{D}^{fc}(k_x, k_y) \vec{C}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z |z-z'|}}{k_z} dk_x dk_y$$

Note change of apices for generality

We know the solution of this integral

$$\vec{f}(x, y, z) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \underbrace{(\bar{D}^{fc}(k_x, k_y) \vec{C}(k_x, k_y))}_{\text{slow varying function in the surrounding of the stationary phase point } k_{xs}, k_{ys}, k_{zs}} \underbrace{\frac{e^{-jk_z |z-z'|}}{k_z} e^{-jk_x x} e^{-jk_y y}}_{\text{rapidly varying function}} dk_x dk_y$$

slow varying function in the surrounding of the **stationary phase point** k_{xs}, k_{ys}, k_{zs}

$$\vec{f}^{far}(x, y, z) = \bar{D}^{fc}(k_{xs}, k_{ys}, z, z') \vec{C}(k_{xs}, k_{ys}) \underbrace{\frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{-jk_z |z-z'|}}{k_z} e^{-jk_x x} e^{-jk_y y} dk_x dk_y}_{je^{-jkr} \over 2\pi r}$$

$$\vec{f}^{far}(\vec{r}) = j \bar{D}^{fc}(k_{xs}, k_{ys}, z, z') \vec{C}(k_{xs}, k_{ys}) \frac{e^{-jkr}}{2\pi r}$$

$$\frac{je^{-jkr}}{2\pi r}$$

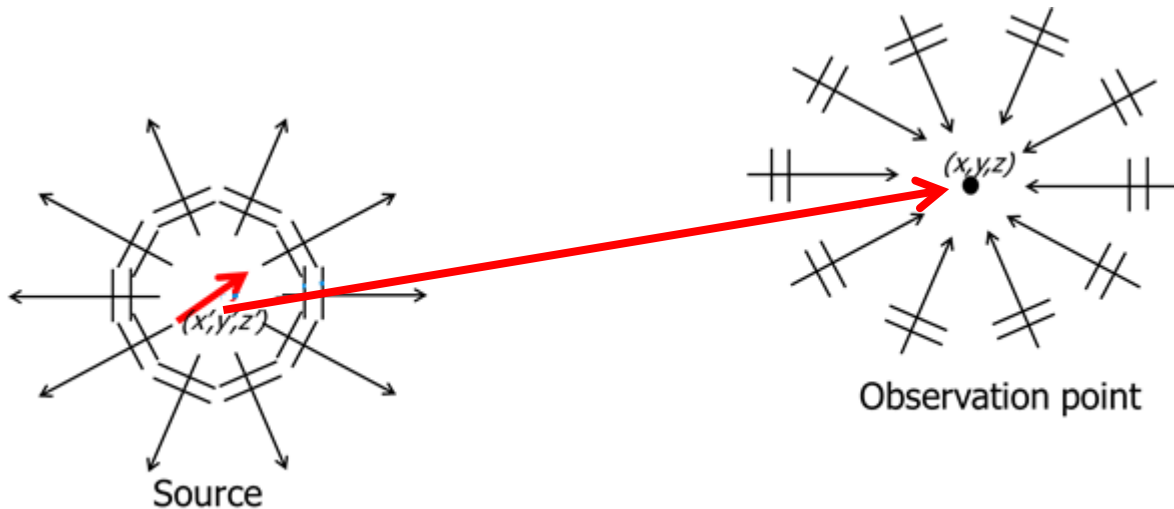
$$r = \sqrt{x^2 + y^2 + (z - z')^2}$$

Dominant PW in the Far Field

$$\vec{f}^{far}(\vec{r}) = j\bar{D}^{fc}(k_{xs}, k_{ys}, z, z')\vec{C}(k_{xs}, k_{ys})\frac{e^{-jkr}}{2\pi r}$$

$$\begin{aligned}x - x' &= r\sin\theta\cos\phi \\y - y' &= r\sin\theta\sin\phi \\z - z' &= r\cos\theta\end{aligned}$$

The far field of any source is proportional to $\frac{e^{-jkr}}{r}$



The plane wave with the dominant role is the one given by the direct ray

$$k_{xs} = k_0\sin\theta\cos\phi$$

$$k_{ys} = k_0\sin\theta\sin\phi$$

$$k_{zs} = k_0\cos\theta$$

It corresponds to a stationary phase point

Far Field Calculation: Stationary Phase Point

$$\vec{E} = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{D}^{fc}(k_x, k_y) \vec{C}(k_x, k_y) e^{-jk_x x} e^{-jk_y y} \frac{e^{-jk_z |z-z'|}}{k_z} dk_x dk_y$$

In order to find a stationary phase point, it is simpler to express the integral into cylindrical integration domain by applying the following change of variables

The observation points are defined by

$$x - x' = r \sin \theta \cos \phi$$

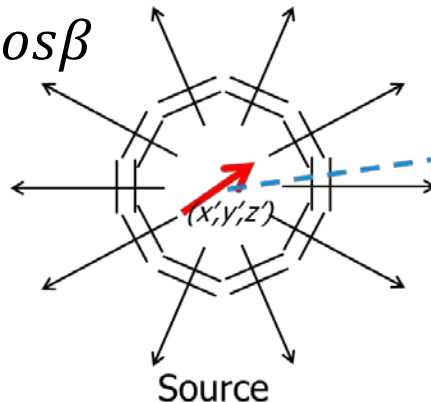
$$y - y' = r \sin \theta \sin \phi$$

$$z - z' = r \cos \theta$$

$$k_x = k_0 \sin \beta \cos \alpha$$

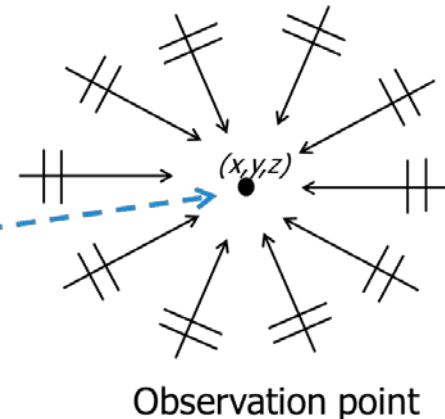
$$k_y = k_0 \sin \beta \sin \alpha$$

$$k_z = k_0 \cos \beta$$



Phase function

$$f(\beta, \alpha) = k_0 r (\sin \beta \cos \alpha \sin \theta \cos \phi + \sin \beta \sin \alpha \sin \theta \sin \phi + \cos \beta |\cos \theta|) = k_0 r (\sin \beta \sin \theta \cos(\alpha - \phi) + \cos \beta |\cos \theta|)$$



Observation point

Stationary Phase Points

$$\left. \frac{\partial f(\beta, \alpha)}{\partial \beta} \right|_{\beta_0, \alpha_0} = \left. \frac{\partial f(\beta, \alpha)}{\partial \alpha} \right|_{\beta_0, \alpha_0} = 0$$

$$f(\beta, \alpha) = k_0 r (\sin \beta \sin \theta \cos(\alpha - \phi) + \cos \beta |\cos \theta|)$$

$$\frac{\partial}{\partial \alpha} f(\beta, \alpha) = -k_0 r \sin \beta \sin \theta \sin(\alpha - \phi) = 0 \quad \longrightarrow \quad \alpha_0 = \phi,$$

$$\frac{\partial}{\partial \beta} f(\beta, \alpha) = k_0 r (\cos \beta \sin \theta \cos(\alpha - \phi) - \sin \beta |\cos \theta|) = 0$$

$$\frac{\partial}{\partial \beta} f(\beta, \alpha_0) = k_0 r (\cos \beta \sin \theta - \sin \beta |\cos \theta|) = 0$$

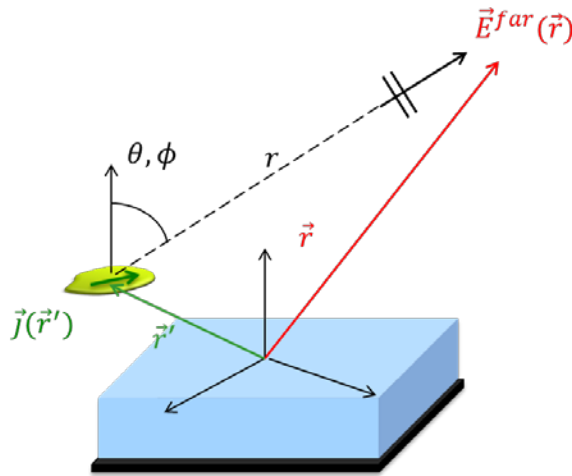
$$z - z' > 0 \quad \longrightarrow \quad \frac{\partial}{\partial \beta} f(\beta, \alpha_0) = k_0 r \sin(\beta - \theta) = 0 \quad \longrightarrow \quad \beta_0 = \theta$$

When this approximation is valid?

$$\vec{f}^{far}(\vec{r}) = j\tilde{G}^{fc}(k_{xs}, k_{ys}, z, z')\vec{C}(k_{xs}, k_{ys})\frac{e^{-jkr}}{2\pi r}$$

Direct PW

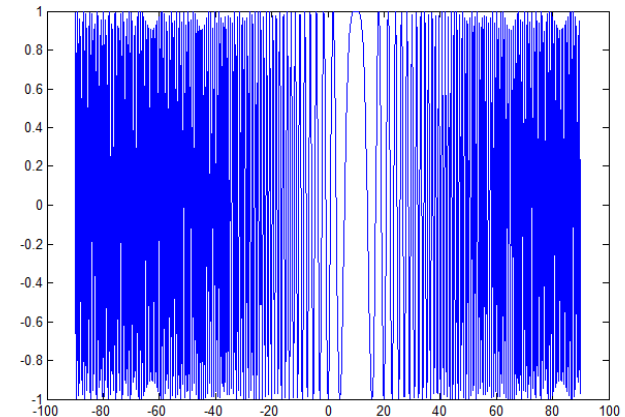
$$k_{xs} = k_0 \sin\theta \cos\phi \quad k_{ys} = k_0 \sin\theta \sin\phi$$



$\vec{C}_0$ has to be
slowly varying

$k_0 r \gg 1$
Highly oscillating
phase

$$r > \frac{2D^2}{\lambda}$$



Truly Important from Lecture 1

$$\frac{e^{-jk|\vec{r}-\vec{r}'|}}{4\pi|\vec{r}-\vec{r}'|} = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} -\frac{e^{-jk_x(x-x')}e^{-jk_y(y-y')}e^{-jk_z(z-z')}}{k^2 - k_x^2 - k_y^2 - k_z^2} dk_x dk_y dk_z$$

Dyadic Free Space Green's functions in spectral domain

$$\bar{\bar{G}}_{fs}^{ej}(\vec{r}-\vec{r}') = \frac{-\zeta}{k8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \begin{bmatrix} k^2 - k_x^2 & -k_x k_y & -k_x(\pm k_z) \\ -k_x k_y & k^2 - k_y^2 & -k_y(\pm k_z) \\ -k_x(\pm k_z) & -k_y(\pm k_z) & k^2 - k_z^2 \end{bmatrix} \frac{e^{-jk_x(x-x')}e^{-jk_y(y-y')}e^{-jk_z|z-z'|}}{\sqrt{k^2 - k_x^2 - k_y^2}} dk_x dk_y$$

$$(z-z') \begin{matrix} > \\ < \end{matrix} 0 \Rightarrow \pm k_z = \sqrt{k^2 - k_x^2 - k_y^2}$$

It will not hurt to know a part of it by heart.

$$\vec{f}^{far}(\vec{r}) = j\bar{\bar{D}}^{fc}(k_{xs}, k_{ys}, z, z') \vec{C}(k_{xs}, k_{ys}) \frac{e^{-jkr}}{2\pi r}$$